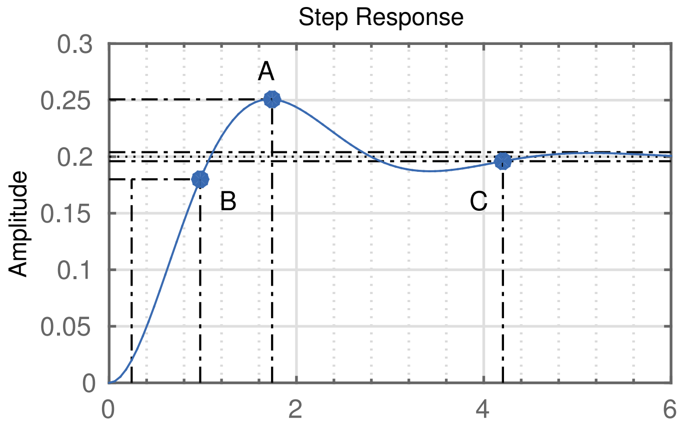
Pre-Laboration Test 5 **Name:** ______________________ **Process:** ___

1. **Which combination is the correct one?**



	A	settling time		A	rise time		A	overshoot		A	overshoot
(a)	B	overshoot	(b)	B	settling time	(c)	B	rise time	(d)	B	settling time
	C	rise time		C	overshoot		C	settling time		C	rise time

2. **Which one is the correct Laplace transform of** $\dot{y} = \dot{x} - 3y + x$

(a) $Y = \frac{s+1}{s+3}X$ (b) $Y = \frac{3}{s^2+s+3}X$ (c) $Y = \frac{1}{s^2+s-3}X$ (d) $Y = \frac{1}{s^2-s+3}X$

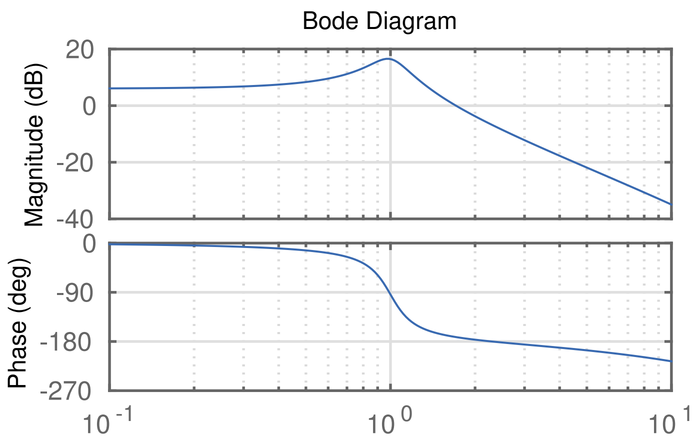
3. **How many zeros does the following transfer function have?**

$$G(s) = \frac{s-8}{s^2-4s-1}$$

(a) 1 (b) 3 (c) 2 (d) 5

4. **What is the phase margin of this system?**

(a) -90°
(b) 15°
(c) -15°
(d) 45°



5. **Which statement is true about feedback control?**
   (a) A stable feedback controller can never render a system unstable.
   (b) You need exact knowledge of the system to build a feedback controller.
   (c) A feedback controller can not stabilize an unstable system.
   (d) A feedback controller can compensate for unknown disturbances.

Answers:

1.	
2.	
3.	
4.	
5.	

Compiled September 23, 2017